PRODUCT USER MANUAL

For High-Resolution WIND Products

WIND_ARC_PHY_HR_L3_NRT_012_100

WIND_ATL_PHY_HR_L3_NRT_012_101

WIND_BAL_PHY_HR_L3_NRT_012_102

WIND_BLK_PHY_HR_L3_NRT_012_103

WIND_MED_PHY_HR_L3_NRT_012_104

WIND_ARC_PHY_HR_L3_MY_012_105

WIND_ATL_PHY_HR_L3_MY_012_106

WIND_BAL_PHY_HR_L3_MY_012_107

WIND_BLK_PHY_HR_L3_MY_012_108

WIND_MED_PHY_HR_L3_MY_012_109

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	31/12/2023	All	Creation of the document	CLS, IFREMER, OCEANSCOPE	Romain Husson
1.1	31/07/2024		MY content added + small updates for NRT	CLS, IFREMER, OCEANSCOPE	Romain Husson
1.2	20/01/2025		Addition of S1B dataset for MY. Addition of the longitude/latitude ancillary variables for ARC product.	Marine De Carlo, Jean-François Piollé	Romain Husson

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
ATL	Eastern North Atlantic (European part of ATLantic Ocean)
BAL	BALtic Sea
BLK	BLacK sea
CDSE	Copernicus Data Space Ecosystem
CF	Climate Forecast (convention for NetCDF)
CLS	Collecte Localisation Satellites
CMEMS	Copernicus Marine Environment Monitoring Service
CRS	Coordinate Reference System
DGF	Directgetfile : service tool (FTP like) to download a NetCDF file
ECMWF	European Center for Medium-Range Weather Forecast
EW	Extended Wide
HR	High-Resolution
IBI	Iberia Biscay Ireland
IFREMER	Institut Français pour la Recherche et l'Exploitation de la MER
IW	Interferometric Wide
MED	MEDiterranean Sea
MFC	Monitoring and Forecasting Centre
MY	Multi-Year
NetCDF	Network Common Data Form
NRT	Near Real-Time
NWS	North-West Shelf Seas
OVL	Ocean Virtual Lab (https://ovl.oceandatalab.com)
PU	Production Unit
PUM	Product User Manual
QUID	QUality Information Document
S1A	Sentinel-1A
S1B	Sentinel-1B
SAR	Synthetic Aperture Radar

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1. INTRODUCTION

1.1. Summary

This document describes data product files from the HR-WINDS Production Unit (PU), what data services are available to access them and how to use the files and services.

The HR-WINDS PU provides High-Resolution (HR) wind products derived from Sentinel-1 at Level-3, i.e. georeferenced derived geophysical variables with associated ancillary information. There are two main production streams in the HR-WINDS PU that both deliver 5 regional products: the near Real-Time (NRT) stream operated by CLS (Collecte Localisation Satellites) and the multi-year (MY) stream operated by IFREMER (Institut Français pour la Recherche et l'Exploitation de la MER).

The products WIND_XXX_PHY_HR_L3_TT_012_10X – where XXX and X correspond to the region name and region number and TT is either 'NRT' or 'MY' – contain Level-3 (L3) gridded high-resolution ocean surface wind observations from Sentinel-1 (S1) SAR data and are distributed by the Copernicus Marine Service.

The 5 regions for which the data are provided are the following CMEMS regions:

- the Arctic Ocean (XXX = ARC; X=0 for NRT, X=5 for MY)
- the Eastern North Atlantic (XXX = ATL; X=1 for NRT, X=6 for MY) including the North-West Shelf Seas (NWS) and the Iberia-Biscay-Ireland seas (IBI)
- the Baltic Sea (XXX = BAL; X=2 for NRT, X=7 for MY)
- the Black Sea (XXX = BLK; X=3 for NRT, X=8 for MY)
- and the Mediterranean Sea (XXX = MED; X=4 for NRT, X=9 for MY)

The L3 products are based on 1-km pixel size S1 Level-2 OCN products produced daily by the European Space Agency (ESA) at the highest available resolution and are gridded with a 0.012 degree resolution (between 0.5 and 1.2 km in zonal direction depending on the latitude and 1.3 km in meridional direction).

To allow concatenation between the dataset, the geographical projection is the same, namely the cylindrical equidistant one, for all regions except for the Arctic Ocean where the stereographic projection is used.

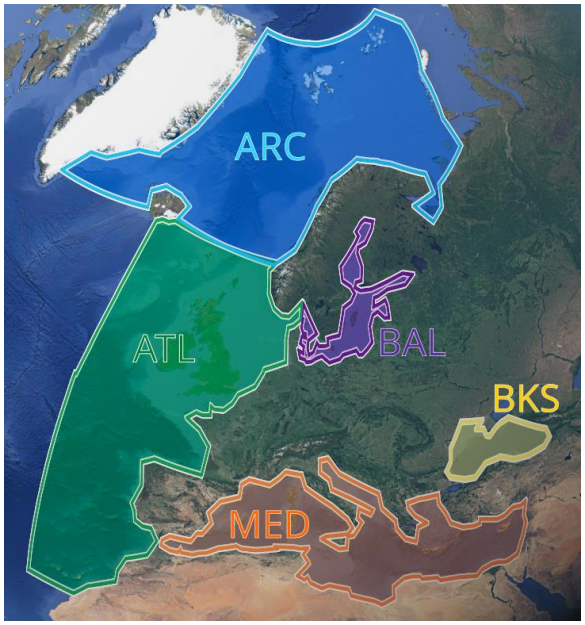


Figure 1: Adapted CMEMS regions

1.2. History of changes

This table describes the changes related to the products and format, while the record table describes the changes related to the document itself.

Date	Description of changes and impacted product	System version (processing & inputs)	Format version
05/04/2024	Creation of the product for NRT products EIS	1.0	1.0
31/07/2024	Update of NRT products for days without any input data	**1.1.0**	**1.0**
	Creation of MY products for EIS With additional flag variables to trace "cosmetic pixels"	**1.1.1**	**1.1**
20/01/2025	Additional ancillary variables longitude/latitude for ARC product Addition of "cosmetic pixels" for NRT (new flag value + new ancillary variable) Addition of S1B datasets for MY	**1.2**	**1.2**

2. DESCRIPTION OF THE PRODUCT SPECIFICATION

2.1. General Information

Product Lines	WIND_ARC_PHY_HR_L3_NRT_012_100 WIND_ATL_PHY_HR_L3_NRT_012_101 WIND_BAL_PHY_HR_L3_NRT_012_102 WIND_BLK_PHY_HR_L3_NRT_012_103 WIND_MED_PHY_HR_L3_NRT_012_104 WIND_ARC_PHY_HR_L3_MY_012_105, WIND_ATL_PHY_HR_L3_MY_012_106, WIND_BAL_PHY_HR_L3_MY_012_107, WIND_BLK_PHY_HR_L3_MY_012_108, WIND_MED_PHY_HR_L3_MY_012_109
Geographical coverage	European coastal areas. See Figure 1
Variables	• Measurement time [s] • Wind speed [m/s] • Wind direction [degrees] • Zonal (eastward) wind velocity [m/s] • Meridional (northward) Wind velocity [m/s] • Model wind speed [m/s] • Model wind direction [degrees] • Model zonal (eastward) wind velocity [m/s] • Model meridional (northward) wind velocity [m/s] • Number_of_pixels [1]
Product Type	Near Real-Time and Multi-Year Level-3 Observations
Available time series	NRT: From 2024/04/01 and on going MY: - S1A: From 2018/03/13 to 2024/12/31 - S1B: From 2018/03/13 to 2021/12/23
Temporal resolution	Daily files with instantaneous measurements
Target delivery time	NRT: 3 hours after the acquisition of upstream data (L2 OCN) MY : Not applicable
Delivery mechanism	Marine Data Store
Horizontal resolution	0.012 deg for ATL, MED, BLK and BAL – 1.3 km for ARC
Number of vertical levels	1 at 10 m above sea level.
Format	NetCDF-4 classic model CF-1.6.

Table 1: HR-Winds L3 products

The High-Resolution Winds L3 products are processed starting from L2 OCN products from ESA's processing stream. The detailed processing is described in the Products' QUality Information Document (QUID).

The observations are gridded either on a cylindrical equidistant projection (for ALT, BLK, BAL and MED) or on a polar stereographic projection (ARC). Each grid point corresponds to one observation at most. For grid points with multiple L2 observations, only the latest one is kept on the L3 product.

The observations can be acquired in multiple polarizations – either horizontal HH+HV or vertical VV+VH – and swath modes – either Interferometric Wide (IW) or Extended Wide (EW). Moreover, they can be processed either with the co-pol or the dual-pol algorithms (more details about these algorithms can be found in the QUID).

The L3 products are daily products that grid and merge all observations regardless of their polarization (Horizontal or Vertical, single-pol or dual-pol), swath mode (IW or EW) or processing (co-pol or dual-pol) algorithm. For each grid point this information is contained in the *product_info* variable.

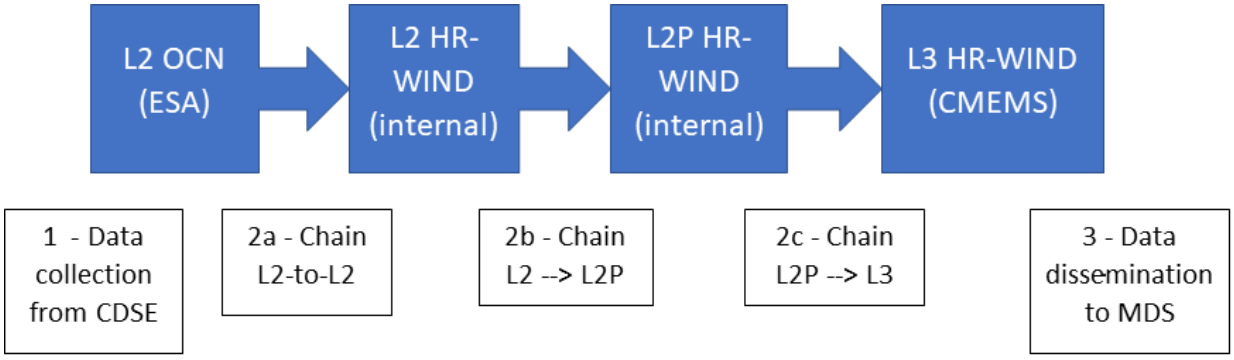


Figure 2: L3 wind processing flowchart.

2.2. Details of datasets

The following tables give details about the datasets that are available in the WIND_XXX_PHY_HR_L3_NRT_012_10X and WIND_XXX_PHY_HR_L3_MY_012_10X products.

Table 2 gives the L3 dataset names that are produced for the different oceanic regions. Note that for each product the ascending and descending passes are separated into different datasets.

Table 3 defines the variable names and units, their corresponding netCDF long_names and CF standard names.

	Region	Datasets
NRT	ARC	cmems_obs-wind_arc_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_arc_phy_nrt_l3-s1a-sar-desc-0.01deg_P1D-i
	ATL	cmems_obs-wind_atl_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_atl_phy_nrt_l3-s1a-sar-desc-0.01deg_P1D-i
	BAL	cmems_obs-wind_bal_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_bal_phy_nrt_l3-s1a-sar-desc-0.01deg_P1D-i
	BLK	cmems_obs-wind_blk_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_blk_phy_nrt_l3-s1a-sar-desc-0.01deg_P1D-i
	MED	cmems_obs-wind_med_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_med_phy_nrt_l3-s1a-sar-desc-0.01deg_P1D-i
MY	ARC	cmems_obs-wind_arc_phy_my_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_arc_phy_my_l3-s1a-sar-desc-0.01deg_P1D-i
		cmems_obs-wind_arc_phy_my_l3-s1b-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_arc_phy_my_l3-s1b-sar-desc-0.01deg_P1D-i
	ATL	cmems_obs-wind_atl_phy_my_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_atl_phy_my_l3-s1a-sar-desc-0.01deg_P1D-i
		cmems_obs-wind_atl_phy_my_l3-s1b-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_atl_phy_my_l3-s1b-sar-desc-0.01deg_P1D-i
	BAL	cmems_obs-wind_bal_phy_my_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_bal_phy_my_l3-s1a-sar-desc-0.01deg_P1D-i
		cmems_obs-wind_bal_phy_my_l3-s1b-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_bal_phy_my_l3-s1b-sar-desc-0.01deg_P1D-i
	BLK	cmems_obs-wind_blk_phy_my_l3-s1a-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_blk_phy_my_l3-s1a-sar-desc-0.01deg_P1D-i
		cmems_obs-wind_blk_phy_my_l3-s1b-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_blk_phy_my_l3-s1b-sar-desc-0.01deg_P1D-i
	MED	cmems_obs-wind_med_phy_my_l3-s1a-sar-asc-0.01deg_P1D-i

		cmems_obs-wind_med_phy_my_l3-s1a-sar-desc-0.01deg_P1D-i
		cmems_obs-wind_med_phy_my_l3-s1b-sar-asc-0.01deg_P1D-i
		cmems_obs-wind_med_phy_my_l3-s1b-sar-desc-0.01deg_P1D-i

Table 2: list of the datasets for the WIND_XXX_PHY_HR_L3_NRT_012_10X and WIND_XXX_PHY_HR_L3_MY_012_10X products

WIND_ARC_PHY_HR_L3_NRT_012_100, WIND_ATL_PHY_HR_L3_NRT_012_101, WIND_BAL_PHY_HR_L3_NRT_012_102, WIND_BLK_PHY_HR_L3_NRT_012_103, WIND_MED_PHY_HR_L3_NRT_012_104	**WIND_ARC_PHY_HR_L3_MY_012_105, WIND_ATL_PHY_HR_L3_MY_012_106, WIND_BAL_PHY_HR_L3_MY_012_107, WIND_BLK_PHY_HR_L3_MY_012_108, WIND_MED_PHY_HR_L3_MY_012_109**
Variables name in the NetCDF file and Unit: Long_name & Standard_name	
time [seconds since 1990-01-01 00:00:00] Validity time time	
For ATL, BAL, BLK and MED	**For ARC**
latitude [degrees_north] latitude latitude	**y** [meters_north] y y
longitue [degrees_east] longitude longitude	**x** [meters_east] x x
	latitude [degrees_north] Ancillary latitude latitude
	longitude [degrees_east] Ancillary longitude longitude
measurement_time [seconds since 1990-01-01 00:00:00] measurement acquisition time time	
crs	
model_wind_speed [m s-1] model wind speed at 10 m wind_speed	
model_wind_to_dir [degree] model wind direction at 10 m wind_to_direction	
wind_speed [m s-1] wind speed at 10 m wind_speed	
wind_to_dir [degree] wind direction at 10 m wind_to_direction	
eastward_wind [m s-1] wind u component at 10 m eastward_wind	
northward_wind [m s-1] wind v component at 10 m northward_wind	
eastward_model_wind [m s-1] model wind u component at 10 m	

eastward_wind
northward_model_wind [m s-1] model wind v component at 10 m northward_wind
product_info [1] Information about the mode, polarization and algorithm used status_flag
quality_level [1] quality level of L3 wind cell aggregate_quality_flag
rejection_flag [1] wind vector reason for rejection status_flag
number_of_pixels [1] number of input L2P pixels used for the L3 reprojected pixel number_of_observations

Table 3: list of the variable names and unit for the WIND_XXX_PHY_HR_L3_TT_012_10X products

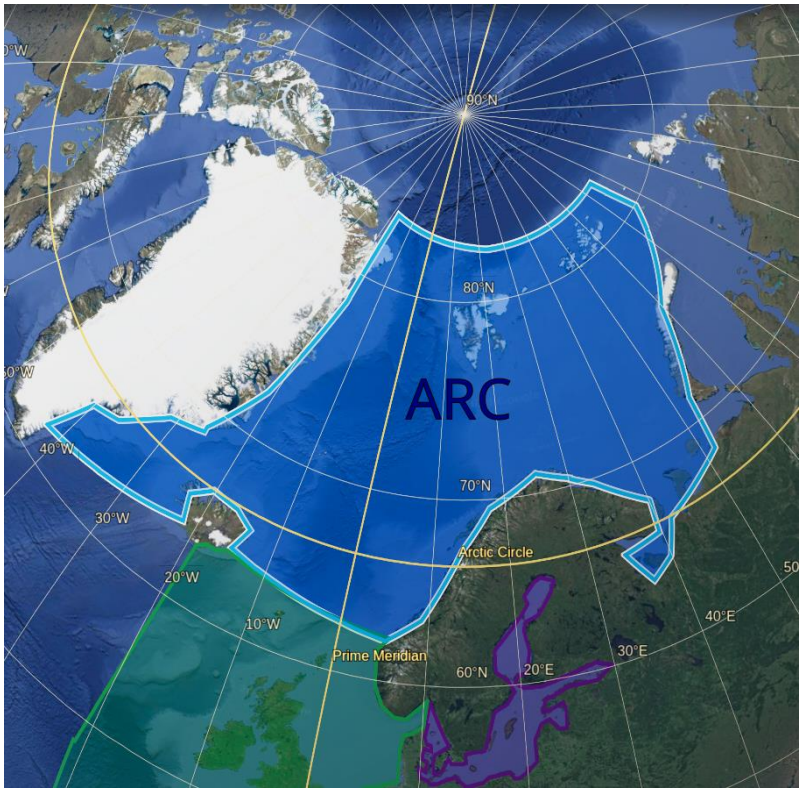
2.3. Additional information

measurement_time [seconds since 1990-01-01 00:00:00]	It is the time at which the satellite measurement was performed. It differs from the variable **time** that contains the generic time of the dataset (no more precise than a julian day).
quality_level	It is a quality flag that indicates whether the data is: - 0 : not existing (no_data) - 1 : bad - 2 : suspect - 3 : acceptable - 4 : good The reason to set this flag below 4 is contained in the **rejection_flag** variable.
rejection_flag	This flag contains various bits explaining the reason for rejection of the observation (see the variable attribute flag_meanings): - some_portion_of_wvc_is_over_land : part of the wind vector cell is over land (taken from L2 OCN) - some_portion_of_wvc_is_over_ice : part of the wind vector cell is over ice (taken from L2 OCN) - not_enough_good_sigma0_for_wind_retrieval: taken from L2 OCN - bright_target_percentage_over_threshold: adapted from L2 OCN - wind_inversion_not_successful: from L2->L3 processing. - cosmetic_pixel (from format version 1.1): pixel added for visualisation interpolating surrounding pixels (nearest neighbour) when the new variable “number_of_pixels” is equal to zero. The association between rejection_flag and quality_level is detailed in the QUID.

`Number_of_pixels`	From format version 1.1 : variable that indicates the number of L2 pixels that are found inside a given L3 pixel. If number_of_pixels = 0, the L3 pixel value is interpolated (nearest-neighbour) and the **cosmetic_pixel** flag meaning of **rejection_flag** is set to 1.
`product_info`	This variable contains 8 flags describing: - The **swath mode** coded in 2 bits: – iw_acq_mode for IW mode, – ew_acq_mode for EW mode - The **emitting polarization** of the observation coded in 2 bits: – v_emitting_pol for vertical polarization, – h_emitting_pol for horizontal polarization - The **receiving polarization** coded in 2 bits: – single_pol_receiving for single polarization, – dual_pol_receiving for dual polarization. - The **algorithm** used coded in 2 bits: – single_pol_wind_processing for co-pol, – dual_pol_wind_processing for dual-pol

Table 4: Description of additional variables

2.4. Production System Description

WIND_ARC_PHY_HR_L3_NRT_012_100 and **WIND_ARC_PHY_HR_L3_MY_012_105**
Grid and resolution: For the Arctic Ocean, the L3 measurements are gridded with a 1.3 km resolution (x/y) using a Polar-stereographic projection: "*Polarstereographic (proj=stere lat_0=90.0 lat_ts=90.0 lon_0=0.0 R=6371000)* " The longitude and latitude variables are given as ancillary data, as a link to the x/y coordinate variables. **Geographic coverage**: This region inherits from the CMEMS Arctic region with a Southern boundary at 63°N. The Northern boundary is adapted to match S1 northmost acquisitions. **Update frequency:** Due to S1 orbit and acquisition plan, the region is sampled on average 3 to 4 times per day by ascending passes and 4 to 5 times by descending passes. => Each dataset contains 3 to 5 S1A or S1B orbits per day. 

WIND_ATL_PHY_HR_L3_NRT_012_101 and **WIND_ATL_PHY_HR_L3_MY_012_106**

Grid and resolution: The grid that is used is a regular lat-lon grid (cylindrical equidistant projection), with a spacing of 0.012 degrees. Therefore, the resolution is between 0.8 and 1.2 km in zonal direction depending on the latitude and 1.3 km in meridional direction.

Geographic coverage: The Eastern North Atlantic (ATL) region corresponds to the concatenation of two CMEMS regions: the North-West Shelf Seas (NWS) and the Iberia-Biscay-Ireland seas (IBI). It extends from 26°N to 62°N in the meridional direction and from 20°W to the coast for the longitudinal direction.

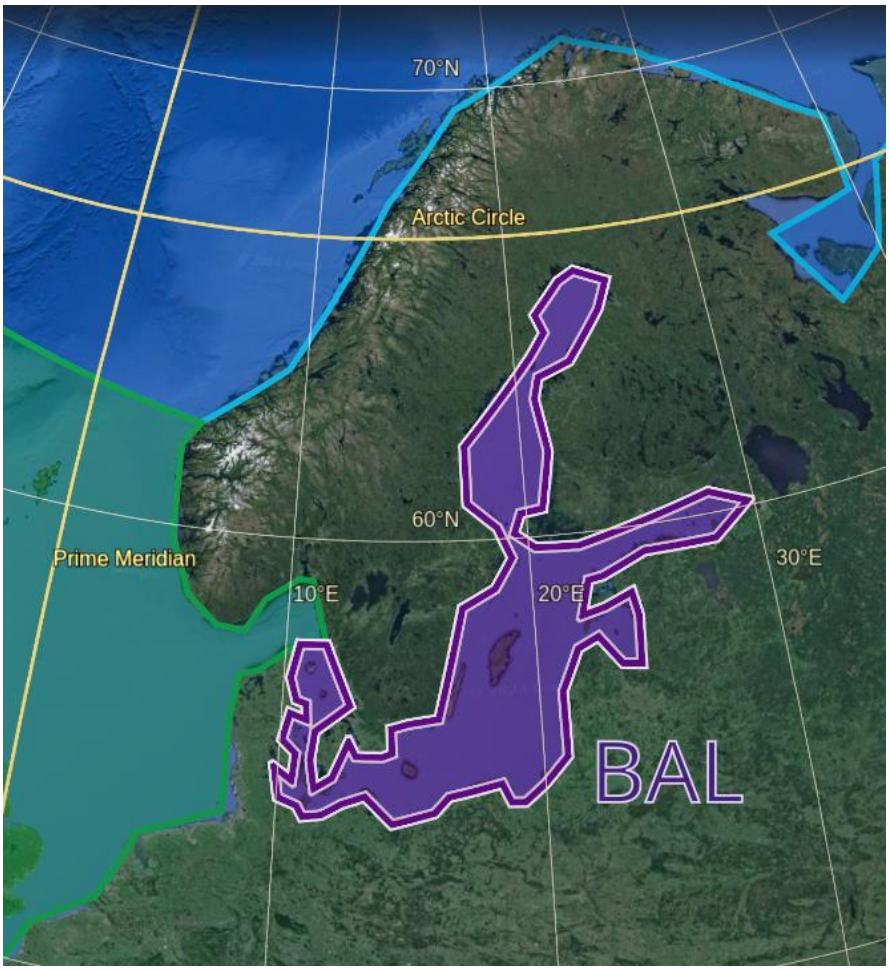
Update frequency: Due to S1 orbit and acquisition plan, the region is sampled 1 to 2 times per day both by ascending passes and by descending passes.
=> Each dataset contains 1 to 2 S1A or S1B orbits per day.

WIND_BAL_PHY_HR_L3_NRT_012_102 and **WIND_BAL_PHY_HR_L3_MY_012_107**

Grid and resolution: The grid that is used is a regular lat-lon grid (cylindrical equidistant projection), with a spacing of 0.012 degrees. Therefore, the resolution is between 0.5 and 0.8 km in zonal direction depending on the latitude and 1.3 km in meridional direction.

Geographic coverage: The Baltic Sea (BAL) region corresponds to the CMEMS BAL region.

Update frequency: Due to S1 orbit and acquisition plan, on average the region is sampled once a day both by ascending passes and by descending passes.
=> Each dataset contains one S1A or S1B orbit per day.

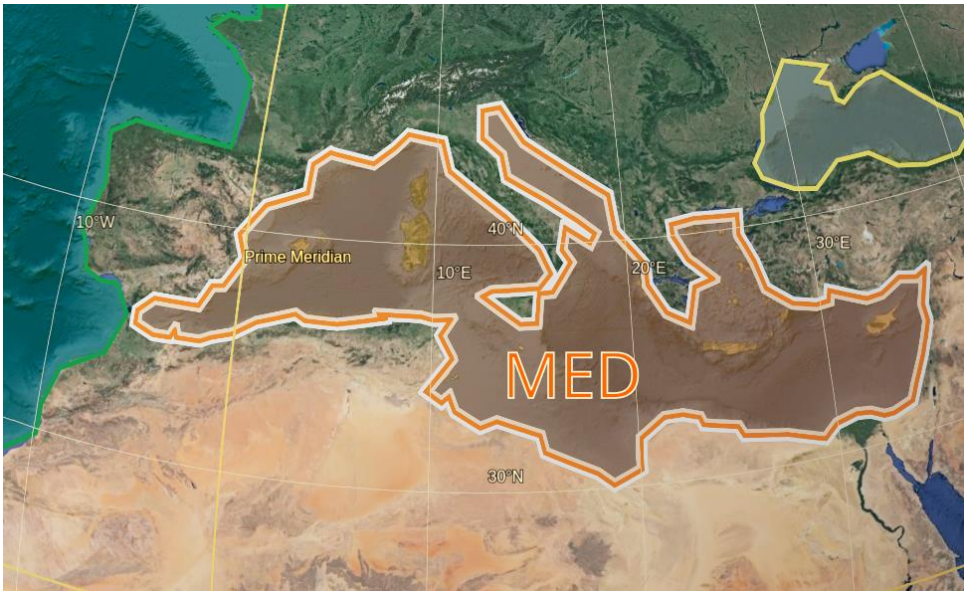


WIND_BLK_PHY_HR_L3_NRT_012_103 and **WIND_BLK_PHY_HR_L3_MY_012_108**

Grid and resolution: The grid that is used is a regular lat-lon grid (cylindrical equidistant projection), with a spacing of 0.012 degrees. Therefore, the resolution is around 1km in zonal direction and 1.3 km in meridional direction.

Geographic coverage: The Black Sea (BLK) region corresponds to the CMEMS BLK region.

Update frequency: Due to S1 orbit and acquisition plan, on average the region is sampled once per day either by an ascending pass or by a descending pass.
=> Each dataset contains at most one S1A or S1B orbit per day

WIND_MED_PHY_HR_L3_NRT_012_104 and **WIND_MED_PHY_HR_L3_MY_012_109**
Grid and resolution: The grid that is used is a regular lat-lon grid (cylindrical equidistant projection), with a spacing of 0.012 degrees. Therefore, the resolution is around 1km in zonal direction and 1.3 km in meridional direction. **Geographic coverage**: The Mediterranean Sea (MED) region corresponds to the CMEMS MED region. **Update frequency:** Due to S1 orbit and acquisition plan, on average the region is sampled twice a day both by ascending passes and by descending passes. => Each dataset contains 2 S1A or S1B orbits per day.   

2.5. Grid

All products except the Arctic one are gridded onto the same cylindrical equidistant projection, with a spacing of 0.012 degrees.

WIND_ARC_PHY_HR_L3_NRT_012_100 and WIND_ARC_PHY_HR_L3_MY_012_105 (the Arctic products) are gridded onto a polar stereographic grid with a 1.3 km resolution.

All the variables are defined at the centre of the output grid cell after spatial averaging.

2.6. Processing information

2.6.1. Update Time

The Near Real-Time and Multi Year products are daily products, for the NRT, the file may be updated several times a day (depending on upstream L2 OCN data).

Every hour, the NRT processing chain checks whether new L2 OCN products are available and the L3 daily product is updated if this is the case (there is an update in the filename, see section.4.2). The target delivery time for the NRT product is 3 hours after the availability of the upstream L2 OCN products. However, best efforts are made to try to make the delivery within 3 hours from the satellite acquisition (including around 50 minutes of processing time from upstream processing). Up to now, the median of total processing time (from acquisition to L3 publication) is around 125 minutes, with 80 % of the data distributed within 3 hours from the acquisition.

Due to the satellite orbit and acquisition plan, the number of updates per day depends on the region considered, it ranges from one update every second day for BLK to four (or five) updates per day for ARC.

2.6.2. Time averaging

There is no time averaging in this product. For L3 grid points that are covered by multiple ascending or descending overpasses during a day, the spatially interpolated value from the overpass with the latest timestamp is included in the L3 product.

2.6.3. Spatial averaging

When more than one L2 pixel is found within a L3 pixel (number_of_pixels >1), the variable value is taken as the average value of those L2 pixels. Regarding the rejection_flag and product_info flag variables , a "bitwise OR" operation is used to compute the output pixel. On the other hand, the quality_level is taken as the minimum of the aggregated quality_level L2 pixels.

3. FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4. FILES NOMENCLATURE

4.1. Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subset Service

Files nomenclature when downloaded through the CMEMS Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnn.nc**

where:

- **datasetname** is a character string within one of the datasets listed in section 2.2
- **nnnnnn**: 6-digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **nc**: standard NetCDF filename extension.

  Example:

  cmems_obs-wind_arc_phy_nrt_l3-s1a-sar-asc-0.01deg_P1D-i_435283.nc

4.2. Nomenclature of original files

The nomenclature used for the WIND_XXX_PHY_HR_L3_TT_012_10X product is:

XXX-HRWIND_**TT**_L3-OBS_S1A_**<res>**_**<A/D>**_**<day>**_**<prod_time>**.nc

Where:

- **XXX** is the geographic zone: ARC, ATL, BAL, BLK and MED.
- **TT** corresponds to the production stream: 'NRT' or 'MY'
- **<res>** = '01', is the L3 pixel size in [degrees * 100]
- **<A/D>** is the orbit orientation: ASC or DES
- **<day>** is the day of measurement with format yyyymmdd.
- **<prod_time>** is the file production timestamp with format yyyymmddTHHMMSS (only for NRT).

Example : ARC-HRWIND_NRT_L3-OBS_S1A_01_ASC_20240401_20240402T034432.nc

4.3. N.B.: The <prod_time> will be updated each time a new L2 OCN pass is available and processed into a new update of the L3 file. File size

The file size varies depending on the region considered in the product. There should be no significant difference in file size between the ascending and descending datasets. The following table shows the order of magnitude of file size expected for each product (average over a 12-days period).

Product	File size expected (MB)
WIND_ARC_PHY_HR_L3_NRT_012_100	110
WIND_ATL_PHY_HR_L3_NRT_012_101	10
WIND_BAL_PHY_HR_L3_NRT_012_102	3
WIND_BLK_PHY_HR_L3_NRT_012_103	2
WIND_MED_PHY_HR_L3_NRT_012_104	5

Table 5: Average file size for various HR-WIND products

4.4 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

5. REFERENCES

The following list is a small selection of scientific papers that could be of interest for the user. More information on how the product is computed (and linked to the following references) can be found in the QUality Information Document (QUID : https://documentation.marine.copernicus.eu/QUID/CMEMS-WIND-QUID-012-100to109.pdf)

Mouche, A., Chapron, B., Zhang, B., & Husson, R. (2017). Combined co- and cross-polarized SAR measurements under extreme wind conditions. IEEE Xplore: IEEE Transactions on Geoscience and Remote Sensing, 55, 6476–6755.

Mouche, A., Chapron, B., Knaff, J., Zhao, Y., Zhang, B., and Combot, C. (2019). Copolarized and Cross-Polarized SAR Measurements for High-Resolution Description of Major Hurricane Wind Structures: Application to Irma Category 5 Hurricane. Journal of Geophysical Research: Oceans, 124(6):3905–3922.

Polverari, F., Portabella, M., Lin, W., Sapp, J. W., Stoffelen, A., Jelenak, Z., & Chang, P. S. (2022). On High and Extreme Wind Calibration Using ASCAT. IEEE Transactions on Geoscience and Remote Sensing, 60, 1–10. https://doi.org/10.1109/TGRS.2021.3079898

Portabella, M., Stoffelen, A., & Johannessen, J. A. (2002). Toward an optimal inversion method for synthetic aperture radar wind retrieval. Journal of Geophysical Research: Oceans, 107(C8), 1-1.

Zhang, B. and Perrie, W. (2012). Cross-Polarized Synthetic Aperture Radar: A New Potential Measurement Technique for Hurricanes. Bulletin of the American Meteorological Society, 93(4):531–541.

Zhang, B., Perrie, W., & He, Y. (2011). Wind speed retrieval from RADARSAT-2 quad-polarization images using a new polarization ratio model. Journal of Geophysical Research: Oceans, 116(C8).

6. ANNEX

The file format used by all datasets is detailed below (as generated by "`ncdump -h`").

N.B.: In these global attributes, there are various version attributes:

- The **`product_version`** corresponds to the version of the generation of the current file. The creation of the file (i.e. corresponding to the first pass) has a `product_version=1.0`. For each update of the file when adding new passes the product_version will be incremented by 0.1 (in the previous example, 3 passes are included so `product_version=1.2`). The first digit will only be incremented if there is a reprocessing.
- The **`format_version`** corresponds to the metadata version of the files. Any change in format version will be documented in the version table (Section 1.2)
- The **`system_version`** is also documented in the version table (Section 1.2) and merges information about the version of the processor L2->L3, the version of L2 OCN products from ESA, the `format_version` of the files and any evolution of the WIND_XXX_PHY_HR_L3_TT_012_10X products.

6.1 Example of NRT netCDF structure

```
netcdf BAL-HRWIND_NRT_L3-OBS_S1A_01_ASC_20250116_20250116T121234 {

dimensions:
        time = UNLIMITED ; // (1 currently)
        latitude = 977 ;
        longitude = 1656 ;

variables:
        double measurement_time(time, latitude, longitude) ;
                measurement_time:_FillValue = 1.e+20 ;
                measurement_time:standard_name = "time" ;
                measurement_time:long_name = "Validity time" ;
                measurement_time:authority = "CF-1.11, ACDD-1.3" ;
                measurement_time:axis = "T" ;
                measurement_time:coverage_content_type = "coordinate" ;
                measurement_time:units = "seconds since 1990-01-01" ;
                measurement_time:calendar = "proleptic_gregorian" ;

        float latitude(latitude) ;
                latitude:_FillValue = NaNf ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
```

```
        latitude:long_name = "latitude" ;
        latitude:authority = "CF-1.11, ACDD-1.3" ;
        latitude:axis = "Y" ;
        latitude:valid_min = -90.f ;
        latitude:valid_max = 90.f ;
        latitude:coverage_content_type = "coordinate" ;

    float longitude(longitude) ;
        longitude:_FillValue = NaNf ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:long_name = "longitude" ;
        longitude:authority = "CF-1.11, ACDD-1.3" ;
        longitude:axis = "X" ;
        longitude:valid_min = 0.f ;
        longitude:valid_max = 360.f ;
        longitude:coverage_content_type = "coordinate" ;

    int64 crs ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:longitude_of_prime_meridian = 0. ;
        crs:semi_major_axis = 6378137. ;
        crs:inverse_flattening = 298.257223563 ;

    short wind_speed(time, latitude, longitude) ;
        wind_speed:_FillValue = -32768s ;
        wind_speed:units = "m s-1" ;
        wind_speed:long_name = "wind speed at 10 m" ;
        wind_speed:binning_method = "mean" ;
        wind_speed:cell_methods = "latitude: longitude: mean (interval: 0.012
                                   degree_N interval: 0.012
                                   degree_E)" ;
        wind_speed:standard_name = "wind_speed" ;
        wind_speed:valid_min = 0s ;
        wind_speed:valid_max = 5000s ;
        wind_speed:coverage_content_type = "physicalMeasurement" ;
        wind_speed:coordinates = "time latitude longitude" ;
        wind_speed:add_offset = 0. ;
        wind_speed:scale_factor = 0.01 ;
        wind_speed:missing_value = -32768s ;

    short wind_to_dir(time, latitude, longitude) ;
        wind_to_dir:_FillValue = -32768s ;
        wind_to_dir:units = "degree" ;
```

```
        wind_to_dir:long_name = "wind direction at 10 m" ;
        wind_to_dir:binning_method = "mean" ;
        wind_to_dir:cell_methods = "latitude: longitude: mean (interval: 0.012
                                    degree_N interval: 0.012
                                    degree_E)" ;
        wind_to_dir:standard_name = "wind_to_direction" ;
        wind_to_dir:valid_min = 0s ;
        wind_to_dir:valid_max = 3600s ;
        wind_to_dir:coverage_content_type = "physicalMeasurement" ;
        wind_to_dir:coordinates = "time latitude longitude" ;
        wind_to_dir:add_offset = 0. ;
        wind_to_dir:scale_factor = 0.1 ;
        wind_to_dir:missing_value = -32768s ;

    short eastward_wind(time, latitude, longitude) ;
        eastward_wind:_FillValue = -32768s ;
        eastward_wind:units = "m s-1" ;
        eastward_wind:binning_method = "mean" ;
        eastward_wind:cell_methods = "latitude: longitude: mean (interval: 0.012
                                    degree_N interval: 0.012
                                    degree_E)" ;
        eastward_wind:long_name = "wind u component at 10 m" ;
        eastward_wind:standard_name = "eastward_wind" ;
        eastward_wind:valid_min = -5000s ;
        eastward_wind:valid_max = 5000s ;
        eastward_wind:coverage_content_type = "physicalMeasurement" ;
        eastward_wind:coordinates = "time latitude longitude" ;
        eastward_wind:add_offset = 0. ;
        eastward_wind:scale_factor = 0.01 ;
        eastward_wind:missing_value = -32768s ;

    short northward_wind(time, latitude, longitude) ;
        northward_wind:_FillValue = -32768s ;
        northward_wind:units = "m s-1" ;
        northward_wind:binning_method = "mean" ;
        northward_wind:cell_methods = "latitude: longitude: mean (interval: 0.012
                                    degree_N interval: 0.012
                                    degree_E)" ;
        northward_wind:long_name = "wind v component at 10 m" ;
        northward_wind:standard_name = "northward_wind" ;
        northward_wind:valid_min = -5000s ;
        northward_wind:valid_max = 5000s ;
        northward_wind:coverage_content_type = "physicalMeasurement" ;
```

```
        northward_wind:coordinates = "time latitude longitude" ;
        northward_wind:add_offset = 0. ;
        northward_wind:scale_factor = 0.01 ;
        northward_wind:missing_value = -32768s ;

    short model_wind_speed(time, latitude, longitude) ;
        model_wind_speed:_FillValue = -32768s ;
        model_wind_speed:units = "m s-1" ;
        model_wind_speed:long_name = "model wind speed at 10 m" ;
        model_wind_speed:standard_name = "wind_speed" ;
        model_wind_speed:binning_method = "mean" ;
        model_wind_speed:cell_methods = "latitude: longitude: mean (interval: 0.012
                                                degree_N    interval:    0.012
                                                degree_E)" ;
        model_wind_speed:valid_min = 0s ;
        model_wind_speed:valid_max = 5000s ;
        model_wind_speed:coverage_content_type = "auxiliaryInformation" ;
        model_wind_speed:background_wind_source = "ECMWF (IFS)" ;
        model_wind_speed:coordinates = "time latitude longitude" ;
        model_wind_speed:add_offset = 0. ;
        model_wind_speed:scale_factor = 0.01 ;
        model_wind_speed:missing_value = -32768s ;

    short model_wind_to_dir(time, latitude, longitude) ;
        model_wind_to_dir:_FillValue = -32768s ;
        model_wind_to_dir:units = "degree" ;
        model_wind_to_dir:long_name = "model wind direction at 10 m" ;
        model_wind_to_dir:standard_name = "wind_to_direction" ;
        model_wind_to_dir:binning_method = "mean" ;
        model_wind_to_dir:cell_methods = "latitude: longitude: mean (interval: 0.012
                                                degree_N    interval:    0.012
                                                degree_E)" ;
        model_wind_to_dir:valid_min = 0s ;
        model_wind_to_dir:valid_max = 3600s ;
        model_wind_to_dir:coverage_content_type = "auxiliaryInformation" ;
        model_wind_to_dir:background_wind_source = "ECMWF (IFS)" ;
        model_wind_to_dir:coordinates = "time latitude longitude" ;
        model_wind_to_dir:add_offset = 0. ;
        model_wind_to_dir:scale_factor = 0.1 ;
        model_wind_to_dir:missing_value = -32768s ;

    ubyte rejection_flag(time, latitude, longitude) ;
        rejection_flag:_FillValue = 0UB ;
        rejection_flag:long_name = "wind vector reason for rejection" ;
```

```
        rejection_flag:standard_name = "status_flag" ;
        rejection_flag:flag_masks = 1b, 2b, 4b, 8b, 16b, 32b ;
        rejection_flag:flag_meanings =
                                "some_portion_of_wvc_is_over_land
                                 some_portion_of_wvc_is_over_ice
                                 not_enough_good_sigma0_for_wind_retrieval
                                 bright_target_percentage_over_threshold
                                 wind_inversion_not_successful
                                 cosmetic_pixel " ;
        rejection_flag:binning_method = "median" ;
        rejection_flag:cell_methods = "latitude: longitude: median (interval: 0.012
                                        degree_N interval: 0.012
                                        degree_E)" ;
        rejection_flag:coverage_content_type = "qualityInformation" ;
        rejection_flag:coordinates = "time latitude longitude" ;

    ubyte product_info(time, latitude, longitude) ;
        product_info:_FillValue = 0UB ;
    string product_info:long_name = "'Information about the mode, polarization and
                                algorithm used'" ;
        product_info:standard_name = "status_flag" ;
        product_info:flag_masks = 1UB, 2UB, 4UB, 8UB, 16UB, 32UB, 64UB, 128UB ;
        product_info:flag_meanings =
                        "iw_acq_mode ew_acq_mode
                        v_emitting_pol h_emitting_pol
                        single_pol_receiving dual_pol_receiving
                        single_pol_wind_processing dual_pol_wind_processing" ;
        product_info:binning_method = "median" ;
        product_info:cell_methods = "latitude: longitude: median (interval: 0.012
                                        degree_N interval: 0.012
                                        degree_E)" ;
        product_info:coordinates = "time latitude longitude" ;

    short eastward_model_wind(time, latitude, longitude) ;
        eastward_model_wind:_FillValue = -32768s ;
        eastward_model_wind:units = "m s-1" ;
        eastward_model_wind:binning_method = "mean" ;
        eastward_model_wind:cell_methods = "latitude: longitude: mean (interval:
                                        0.012 degree_N interval: 0.012
                                        degree_E)" ;
        eastward_model_wind:long_name = "model wind u component at 10 m" ;
        eastward_model_wind:standard_name = "eastward_wind" ;
        eastward_model_wind:valid_min = -5000s ;
```

```
        eastward_model_wind:valid_max = 5000s ;
        eastward_model_wind:coverage_content_type = "auxiliaryInformation" ;
        eastward_model_wind:background_wind_source = "ECMWF (IFS)" ;
        eastward_model_wind:coordinates = "time latitude longitude" ;
        eastward_model_wind:add_offset = 0. ;
        eastward_model_wind:scale_factor = 0.01 ;
        eastward_model_wind:missing_value = -32768s ;

    short northward_model_wind(time, latitude, longitude) ;
        northward_model_wind:_FillValue = -32768s ;
        northward_model_wind:units = "m s-1" ;
        northward_model_wind:binning_method = "mean" ;
        northward_model_wind:cell_methods = "latitude: longitude: mean (interval: 0.012 degree_N interval: 0.012 degree_E)" ;
        northward_model_wind:long_name = "model wind v component at 10 m" ;
        northward_model_wind:standard_name = "northward_wind" ;
        northward_model_wind:valid_min = -5000s ;
        northward_model_wind:valid_max = 5000s ;
        northward_model_wind:coverage_content_type = "auxiliaryInformation" ;
        northward_model_wind:background_wind_source = "ECMWF (IFS)" ;
        northward_model_wind:coordinates = "time latitude longitude" ;
        northward_model_wind:add_offset = 0. ;
        northward_model_wind:scale_factor = 0.01 ;
        northward_model_wind:missing_value = -32768s ;

    byte quality_level(time, latitude, longitude) ;
        quality_level:_FillValue = 0b ;
        quality_level:long_name = "quality level of L3 wind cell" ;
        quality_level:flag_values = "[0 1 2 3 4]" ;
        quality_level:flag_meanings = "no_data bad suspect acceptable good" ;
        quality_level:binning_method = "median" ;
        quality_level:cell_methods = "latitude: longitude: median (interval: 0.012 degree_N interval: 0.012 degree_E)" ;
        quality_level:standard_name = "aggregate_quality_flag" ;
    string quality_level:comment = "These are the overall quality indicators, the reason for classifying <4 is given in 'rejection_flag'" ;
        quality_level:coverage_content_type = "qualityInformation" ;
        quality_level:coordinates = "time latitude longitude" ;

    double number_of_pixel(time, latitude, longitude) ;
        number_of_pixel:_FillValue = NaN ;
```

```
        number_of_pixel:binning_method = "count" ;
        number_of_pixel:units = "1" ;
        number_of_pixel:long_name = "number of input L2P pixels used for the L3
                                     reprojected pixel" ;


                                     number_of_pixel:standard_name =
                                     "number_of_observation" ;

    int64 time(time) ;
        time:units = "days since 2024-04-01 00:00:00" ;
        time:calendar = "proleptic_gregorian" ;


// global attributes:
    :geospatial_lat_min = 53.95 ;
    :geospatial_lat_max = 65.67 ;
    :geospatial_lon_min = 9.85 ;
    :geospatial_lon_max = 29.72 ;
    :spatial_resolution = "0.012 degree" ;
    :temporal_resolution = "daily" ;
    :geospatial_lat_resolution = "0.012 degree" ;
    :geospatial_lon_resolution = "0.012 degree" ;
    :processing_level = "L3" ;
    :geospatial_bounds = "POLYGON((53.956 65.66800000000045, 53.956
                          29.716000000000754, 9.856 29.716000000000754,
                          9.856 65.66800000000045, 53.956
                          65.66800000000045))" ;
    :geospatial_bounds_crs = "EPSG:4326" ;
    :cdm_data_type = "grid" ;
    :title = "Baltic Sea - Wind - SAR S1A - 1km daily Ascending" ;
    :summary = "This dataset provides L3 gridded wind observations obtained from S1A SAR
                data inversion." ;
    :id = "IFR-L3C-BAL-L2_OCN" ;
    :time_coverage_start = "2025-01-16T00:00:00Z" ;
    :time_coverage_end = "2025-01-16T23:59:59Z" ;
    :input_granules = "" ;
    :cmems_product_id = "WIND_BAL_PHY_HR_L3_NRT_012_102" ;
    :naming_authority = "cmems" ;
```

:references = "Product User Manual For L3 HR-Wind Products, v1.2, Jan. 2025, https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf" ;
:description = "Daily concatenation of all S1 Ascending passes over the region" ;
:keywords = "Oceans > Ocean Wind > SAR data" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:metadata_link = "https://resources.marine.copernicus.eu/product-detail/WIND_BAL_PHY_HR_L3_NRT_012_102/INFORMATION" ;
:date_metadata_modified = "2025-01-20-T00:00:00Z" ;
:metadata_conventions = "Climate and Forecast (CF) 1.7, Attribute Convention for Data Discovery (ACDD) 1.3" ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: this dataset is produced by the HR-Wind Production Unit, at CLS, Plouzane (France): It is funded and distributed by Copernicus Marine Service" ;
:Conventions = "CF-1.7, ACDD-1.3, ISO 8601" ;
:standard_name_vocabulary = "Climate and Forecast (CF) Standard Name Table v79" ;
:publisher_institution = "Copernicus Marine Service" ;
:publisher_name = "Copernicus Marine Service" ;
:publisher_url = "https://marine.copernicus.eu" ;
:project = "HR-Wind, Copernicus Marine Service (CMEMS)" ;
:license = "These data are available free of charge under the CMEMS data policy, refer to http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/" ;
:creator_institution = "Collecte Localisation Satellites" ;
:creator_name = "HR-Wind NRT Production Unit" ;
:technical_support_contact = "servicedesk.cmems@groupcls.com" ;
:scientific_support_contact = "rhusson@groupcls.com" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_bounds_vertical_crs = "EPSG:5831" ;
:time_coverage_duration = "P1D" ;
:platform = "Sentinel-1A" ;
:instrument = "C-Band SAR" ;
:band = "C" ;
:instrument_type = "synthetic_aperture_radar" ;
:instrument_vocabulary = "CEOS instrument, https://database.eohandbook.com/database/instrumentindex.aspx" ;
:platform_vocabulary = "CEOS mission, https://database.eohandbook.com/database/missionindex.aspx" ;
:orbit_orientation = "Ascending" ;
:timeliness = "NRT" ;

```
        :processing_software = "Telemachus 3.1.2, SARTool-wind 2.3.7" ;
        :source = "HR-Wind" ;
        :input_dataset = "L2 OCN Sentinel-1A NRT" ;
        :comment = "All wind directions in oceanographic convention (0 deg. flowing North)" ;
        :uuid = "101ae950-b839-11ee-81de-0cc47a3f72ab" ;
        :netcdf_version_id = "4.9.2 of Sep 1 2023 15:18:59" ;
        :format_version = "1.2" ;
        :system_version = "1.2" ;
        :date_created = "2025-01-16T12:12:34Z" ;
        :date_modified = "2025-01-16T12:12:34Z" ;
        :date_issued = "2025-01-16T12:12:34Z" ;
        :product_version = "1.0" ;
        :history = " BAL-HRWIND_NRT_L3-OBS_S1A_01_ASC_20250116_20250116T121234.nc"
;
        :doi = "https://doi.org/10.48670/mds-00332" ;
    }
```

Example of MY netCDF structure

```
netcdf ATL-HRWIND_MY_L3-OBS_S1A_1_ASC_20230101_20241003T114710 {
dimensions:
    time = UNLIMITED ; // (1 currently)
    latitude = 3176 ;
    longitude = 2623 ;
variables:
    int measurement_time(time, latitude, longitude) ;
        measurement_time:long_name = "Validity time" ;
        measurement_time:standard_name = "time" ;
        measurement_time:authority = "CF-1.11, ACDD-1.3" ;
        measurement_time:axis = "T" ;
        measurement_time:coverage_content_type = "coordinate" ;
        measurement_time:coordinates = "time latitude longitude" ;
        measurement_time:units = "seconds since 1990-01-01" ;
        measurement_time:calendar = "proleptic_gregorian" ;
    float latitude(latitude) ;
```

```
        latitude:_FillValue = NaNf ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:long_name = "latitude" ;
        latitude:authority = "CF-1.11, ACDD-1.3" ;
        latitude:axis = "Y" ;
        latitude:valid_min = -90.f ;
        latitude:valid_max = 90.f ;
        latitude:coverage_content_type = "coordinate" ;
    float longitude(longitude) ;
        longitude:_FillValue = NaNf ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:long_name = "longitude" ;
        longitude:authority = "CF-1.11, ACDD-1.3" ;
        longitude:axis = "X" ;
        longitude:valid_min = 0.f ;
        longitude:valid_max = 360.f ;
        longitude:coverage_content_type = "coordinate" ;
    int64 crs ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:longitude_of_prime_meridian = 0. ;
        crs:semi_major_axis = 6378137. ;
        crs:inverse_flattening = 298.257223563 ;
    short wind_speed(time, latitude, longitude) ;
        wind_speed:_FillValue = -32768s ;
        wind_speed:binning_method = "mean" ;
        wind_speed:cell_methods = "latitude: longitude: mean (interval: 0.011996221662468966 degree_N
                    interval: 0.011995425085780095 degree_E)" ;
        wind_speed:long_name = "wind speed at 10 m" ;
        wind_speed:standard_name = "wind_speed" ;
        wind_speed:valid_min = 0s ;
        wind_speed:valid_max = 5000s ;
        wind_speed:units = "m s-1" ;
        wind_speed:coverage_content_type = "physicalMeasurement" ;
        wind_speed:coordinates = "time latitude longitude" ;
        wind_speed:add_offset = 0. ;
        wind_speed:scale_factor = 0.01 ;
        wind_speed:missing_value = -32768s ;
    short wind_to_dir(time, latitude, longitude) ;
        wind_to_dir:_FillValue = -32768s ;
```

```
    wind_to_dir:binning_method = "mean" ;
    wind_to_dir:cell_methods = "latitude: longitude: mean (interval: 0.011996221662468966 degree_N
            interval: 0.011995425085780095 degree_E)" ;
    wind_to_dir:standard_name = "wind_to_direction" ;
    wind_to_dir:long_name = "wind direction at 10 m" ;
    wind_to_dir:units = "degree" ;
    wind_to_dir:valid_min = 0s ;
    wind_to_dir:valid_max = 3600s ;
    wind_to_dir:coverage_content_type = "physicalMeasurement" ;
    wind_to_dir:coordinates = "time latitude longitude" ;
    wind_to_dir:add_offset = 0. ;
    wind_to_dir:scale_factor = 0.1 ;
    wind_to_dir:missing_value = -32768s ;
short model_wind_speed(time, latitude, longitude) ;
    model_wind_speed:_FillValue = -32768s ;
    model_wind_speed:units = "m s-1" ;
    model_wind_speed:long_name = "model wind speed at 10 m" ;
    model_wind_speed:standard_name = "wind_speed" ;
    model_wind_speed:binning_method = "mean" ;
    model_wind_speed:cell_methods = "latitude: longitude: mean (interval: 0.011996221662468966
            degree_N interval: 0.011995425085780095 degree_E)" ;
    model_wind_speed:valid_min = 0s ;
    model_wind_speed:valid_max = 5000s ;
    model_wind_speed:coverage_content_type = "auxiliaryInformation" ;
    model_wind_speed:background_wind_source = "ECMWF (IFS)" ;
    model_wind_speed:coordinates = "time latitude longitude" ;
    model_wind_speed:add_offset = 0. ;
    model_wind_speed:scale_factor = 0.01 ;
    model_wind_speed:missing_value = -32768s ;
short model_wind_to_dir(time, latitude, longitude) ;
    model_wind_to_dir:_FillValue = -32768s ;
    model_wind_to_dir:units = "degree" ;
    model_wind_to_dir:long_name = "model wind direction at 10 m" ;
    model_wind_to_dir:standard_name = "wind_to_direction" ;
    model_wind_to_dir:binning_method = "mean" ;
    model_wind_to_dir:cell_methods = "latitude: longitude: mean (interval: 0.011996221662468966
            degree_N interval: 0.011995425085780095 degree_E)" ;
    model_wind_to_dir:valid_min = 0s ;
    model_wind_to_dir:valid_max = 3600s ;
    model_wind_to_dir:coverage_content_type = "auxiliaryInformation" ;
    model_wind_to_dir:background_wind_source = "ECMWF (IFS)" ;
```

```
        model_wind_to_dir:coordinates = "time latitude longitude" ;
        model_wind_to_dir:add_offset = 0. ;
        model_wind_to_dir:scale_factor = 0.1 ;
        model_wind_to_dir:missing_value = -32768s ;
    ubyte rejection_flag(time, latitude, longitude) ;
        rejection_flag:_FillValue = 0UB ;
        rejection_flag:standard_name = "status_flag" ;
        rejection_flag:long_name = "wind vector reason for rejection" ;
        rejection_flag:flag_masks = 1b, 2b, 4b, 8b, 16b 32b;
        rejection_flag:flag_meanings = "some_portion_of_wvc_is_over_land
                some_portion_of_wvc_is_over_ice not_enough_good_sigma0_for_wind_retrieval
                bright_target_percentage_over_threshold wind_inversion_not_successful
                cosmetic_pixel" ;
        rejection_flag:coverage_content_type = "qualityInformation" ;
        rejection_flag:cell_methods = "latitude: longitude: bitwise_or (interval: 0.011996221662468966
                degree_N interval: 0.011995425085780095 degree_E)" ;
        rejection_flag:binning_method = "bitwise_or" ;
        rejection_flag:coordinates = "time latitude longitude" ;
    ubyte product_info(time, latitude, longitude) ;
        product_info:_FillValue = 0UB ;
        product_info:standard_name = "status_flag" ;
    string product_info:long_name = "Information about the mode, polarization and algorithm used" ;
        product_info:flag_masks = 1UB, 2UB, 4UB, 8UB, 16UB, 32UB, 64UB, 128UB ;
    product_info:flag_meanings = "iw_acq_mode ew_acq_mode v_emitting_pol h_emitting_pol
                single_pol_receiving dual_pol_receiving single_pol_wind_processing
                dual_pol_wind_processing" ;
        product_info:coverage_content_type = "qualityInformation" ;
        product_info:cell_methods = "latitude: longitude: bitwise_or (interval: 0.011996221662468966
                degree_N interval: 0.011995425085780095 degree_E)" ;
        product_info:binning_method = "bitwise_or" ;
        product_info:coordinates = "time latitude longitude" ;
    byte quality_level(time, latitude, longitude) ;
        quality_level:_FillValue = 0b ;
        quality_level:standard_name = "aggregate_quality_flag" ;
        quality_level:long_name = "quality level of L3 wind cell" ;
        quality_level:flag_masks = 1b, 2b, 4b, 8b, 16b ;
        quality_level:flag_meanings = "no_data bad suspect acceptable good" ;
        quality_level:coverage_content_type = "qualityInformation" ;
        quality_level:cell_methods = "latitude: longitude: bitwise_or (interval: 0.011996221662468966
                degree_N interval: 0.011995425085780095 degree_E)" ;
        quality_level:binning_method = "bitwise_or" ;
```

```
    string quality_level:comment = "These are the overall quality indicators, the reason for classifying <4  is
                given in 'rejection_flag'" ;
        quality_level:flag_values = "[0 1 2 3 4]" ;
        quality_level:coordinates = "time latitude longitude" ;
    double number_of_pixel(time, latitude, longitude) ;
        number_of_pixel:_FillValue = NaN ;
        number_of_pixel:binning_method = "count" ;
        number_of_pixel:units = "1" ;
        number_of_pixel:long_name = "number of input L2P pixels used for the L3 reprojected pixel" ;
        number_of_pixel:standard_name = "number_of_observation" ;
    int64 time(time) ;
        time:units = "days since 1990-01-01" ;
        time:calendar = "proleptic_gregorian" ;

// global attributes:
        :geospatial_lat_min = 26.0135 ;
        :geospatial_lat_max = 64.12 ;
        :geospatial_lon_min = -21.0971 ;
        :geospatial_lon_max = 10.37 ;
        :spatial_resolution = "0.012 degree" ;
        :temporal_resolution = "daily" ;
        :geospatial_lat_resolution = "0.012 degree" ;
        :geospatial_lon_resolution = "0.012 degree" ;
        :processing_level = "L3" ;
        :geospatial_bounds = "POLYGON((26.0195 64.11950000000144, 26.0195 10.372900000001192, -
                21.0911 10.372900000001192, -21.0911 64.11950000000144, 26.0195
                64.11950000000144))" ;
        :geospatial_bounds_crs = "EPSG:4326" ;
        :cdm_data_type = "grid" ;
        :title = "Atlantic Ocean - Wind - SAR S1A - 1km daily Ascending" ;
        :summary = "This dataset provides L3 gridded wind observations obtained from S1A SAR data
                inversion." ;
        :id = "IFR-L3C-ATL-L2_OCN" ;
        :time_coverage_start = "2023-01-01T00:00:00Z" ;
        :time_coverage_end = "2023-01-02T00:00:00Z" ;
        :input_granules = " s1a-iw-owi-dv-20230101t165342-20230101t165413-046593-059574.nc, s1a-iw-
                owi-dv-20230101t165407-20230101t165438-046593-059574.nc, s1a-iw-owi-
                dv-20230101t165549-20230101t165617-046593-059574.nc s1a-iw-owi-dv-
                20230101t182623-20230101t182654-046594-05957C.nc s1a-iw-owi-dv-
                20230101t182648-20230101t182718-046594-05957C.nc s1a-iw-owi-dv-
                20230101t182805-20230101t182833-046594-05957C.nc s1a-iw-owi-dv-
```

20230101t182830-20230101t182858-046594-05957C.nc s1a-iw-owi-dv-
20230101t182855-20230101t182925-046594-05957C.nc s1a-iw-owi-dv-
20230101t182919-20230101t182950-046594-05957C.nc s1a-iw-owi-dv-
20230101t182944-20230101t183015-046594-05957C.nc s1a-iw-owi-dv-
20230101t183009-20230101t183040-046594-05957C.nc s1a-iw-owi-dv-
20230101t183034-20230101t183105-046594-05957C.nc s1a-iw-owi-dv-
20230101t183059-20230101t183129-046594-05957C.nc s1a-iw-owi-dv-
20230101t183124-20230101t183154-046594-05957C.nc s1a-iw-owi-dv-
20230101t183148-20230101t183219-046594-05957C.nc s1a-iw-owi-dv-
20230101t183213-20230101t183244-046594-05957C.nc s1a-iw-owi-dv-
20230101t183238-20230101t183309-046594-05957C.nc s1a-iw-owi-dv-
20230101t183303-20230101t183333-046594-05957C.nc s1a-iw-owi-dv-
20230101t183328-20230101t183352-046594-05957C.nc s1a-iw-owi-dv-
20230101t183417-20230101t183448-046594-05957D.nc s1a-iw-owi-dv-
20230101t183442-20230101t183513-046594-05957D.nc" ;

:cmems_product_id = "WIND_ATL_PHY_HR_L3_MY_012_106" ;
:naming_authority = "cmems" ;
:references = 'Product User Manual For L3 HR-Wind Products, v1.2, January 2025, https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WIND-PUM-012-100to109.pdf';
:description = "Daily concatenation of all S1 Ascending passes over the region" ;
:keywords = "Oceans > Ocean Wind > SAR data" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords";
:comment = "All wind directions in oceanographic convention (0 deg. flowing North)" ;
:metadata_link = "https://resources.marine.copernicus.eu/product-detail/WIND_ATL_PHY_HR_L3_MY_012_106/INFORMATION" ;
:date_metadata_modified = "2025-01-20T00:00:00Z" ;
:metadata_conventions = "Climate and Forecast (CF) 1.7, Attribute Convention for Data Discovery (ACDD) 1.3" ;
:doi = "https://doi.org/10.48670/mds-00336" ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: this dataset is produced by the HR-Wind Production Unit, at IFREMER, Plouzane (France): It is funded and distributed by Copernicus Marine Service" ;
:Conventions = "CF-1.7, ACDD-1.3, ISO 8601" ;
:standard_name_vocabulary = "Climate and Forecast (CF) Standard Name Table v79" ;
:publisher_institution = "Copernicus Marine Service" ;
:publisher_name = "Copernicus Marine Service" ;
:publisher_url = "https://marine.copernicus.eu" ;
:project = "HR_Wind, Copernicus Marine Service (CMEMS)" ;
:license = "These data are available free of charge under the CMEMS data policy, refer to http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/";

```
        :creator_institution = “IFREMER" ;
        :creator_name = "HR-Wind Production Unit" ;
        :technical_support_contact = "cersat@ifremer.fr" ;
        :scientific_support_contact = "amouche@ifremer.fr" ;
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_units = "degrees_east" ;
        :geospatial_bounds_vertical_crs = "EPSG:5831" ;
        :time_coverage_duration = "P1D" ;
        :platform = "Sentinel-1 A" ;
        :instrument = "C-Band SAR" ;
        :band = "C" ;
        :instrument_type = "synthetic_aperture_radar" ;
        :instrument_vocabulary = "CEOS instrument,
                https://database.eohandbook.com/database/instrumentindex.aspx" ;
        :platform_vocabulary = "CEOS mission,
                https://database.eohandbook.com/database/missionindex.aspx" ;
        :orbit_orientation = "Ascending" ;
        :timeliness = "MY" ;
        :based_on = "L2 OCN Sentinel-1A NRT" ;
        :source = "HR-Wind" ;
        :input_dataset = "L2 OCN Sentinel-1A NRT" ;
        :uuid = "020042df-9f4c-44be-af1c-5bf18e052b85" ;
        :product_version = "1.0" ;
        :format_version = "1.2" ;
        :system_version = "1.2 ;
        :time_coverage_stop = "2023-01-01T23:59:59Z" ;
        :date_created = "2025-01-23T11:47:10Z" ;
        :date_modified = "2025-01-23T11:47:10Z" ;
        :date_issued = "2025-01-23T11:47:10Z" ;
        :history = "ATL-HRWIND_MY_L3-OBS_S1A_1_ASC_20230101_20241013T114710.nc" ;
        :processing_software = "Telemachus 3.1.2, sar_l3_hrwind 0.1.7 ;
        :netcdf_version_id = "4.9.2" ;
}
```